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Studio Session	Friday 12:00pm

Part A: URL of the Visualisation

Paste your visualisation URL in the empty space below.

Tip: Test your URL in incognito mode to ensure it is a working URL

<https://kbeh0007.github.io/A2/>

Part B: Domain Description

Describe your domain chosen for DV II

The domain is Formula 1 motor racing, a global sport with 75 years of competitive history, rich statistical records, and a Malaysian chapter through the Sepang International Circuit. F1 operates across engineering, economics, geography, and national identity, making it a data-rich domain where visualisation can surface patterns that raw statistics alone cannot reveal.

Describe the Why and Who aspects of your DV II

The central question is: what does F1's history tell us about Malaysia's place in the sport, and can Malaysia justify its return? The task is explanatory, moving from global context through national focus to a forward-looking economic argument.

The intended audience is F1 fans, Malaysian sports policy stakeholders, and data-literate readers interested in F1's growth trajectory and the economics of hosting. Tooltips, pull-quotes, and inline annotations guide readers through each chart's key insight.

Describe the “What” aspect of your DV II

The primary dataset is the Kaggle F1 World Championship Dataset (Rohan Rao, 2021), containing race results, qualifying times, driver standings, constructor records, and circuit metadata. This was supplemented by Formulapedia F1 Statistics for constructor win counts, GPDestinations for attendance and ticket prices, The World Data for F1 revenue and TV viewership, Wikipedia for Malaysian Grand Prix event facts, and Malaysia's international tourism receipts (1995–2020: World Bank ST.INT.RCPT.CD; 2021–2024: DOSM Tourism Satellite Account). Data was aggregated and joined using Vega-Lite's built-in transforms — aggregate, joinaggregate, lookup, calculate, filter, and window — without external preprocessing.

Describe the “How” aspect of your DV II

Idioms were chosen by matching data type to the most effective mark and channel. Position encodes ordered quantitative attributes wherever possible, as it yields the lowest perceptual estimation error. Where position was consumed by geographic coordinates, size and colour luminance were used instead. Using two channels redundantly for one attribute, as in the proportional symbol maps, improves robustness when marks overlap in dense geographic regions like Europe.

The lollipop chart uses position as the primary ordered channel, with colour hue encoding categorical podium rank. The heatmap uses a sequential luminance scale across a two-key constructor-by-decade table, where lighter red encodes fewer wins and darker red encodes dominance. The stacked bar chart normalises win share by season total using a joinaggregate transform, making eras with different race counts directly comparable. Categorical team colours are applied consistently across the Sepang connected dot plot and the constructor win share chart, allowing viewers to transfer colour knowledge between sections.

The Sepang connected dot plot combines rule, circle, and square marks in one layered idiom to simultaneously encode grid position, finish position, and winning constructor per race year. The 2021 bump chart layers translucent area ribbons beneath line marks using separate filtered layers, encoding the championship gap between title contenders visually rather than numerically. The Silverstone qualifying line chart overlays three coloured era band rectangles with a vertical rule annotation to explain why lap times rose permanently from 2010. The conclusion area chart combines an annotated horizontal rule marking the F1 hosting fee against Malaysia's full tourism receipts recovery trajectory, making the economic affordability argument explicit through the data itself rather than through text alone.

Part C: Sketch

A URL to a PDF document with an image of your sketch on your GitHub account.

Tip: Test your URL in incognito mode to ensure it is a working URL

<https://github.com/kbeh0007/A2/blob/e2910177b5539ab18e1d96d27d433ee7365338e1/FIT2179%20DV2%20Sketch.pdf>

Paste a screenshot of your sketch for your DV II.

Note: Use the entire page here for the screenshot and ensure that it is clearly legible and visible

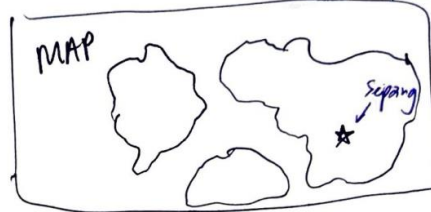
F1 The World at Full Speed

Beh Kai Ying
36297135

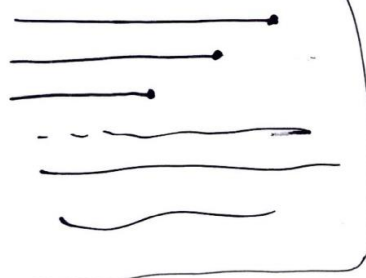
XX Seasons X+ Fans \$3B Revenue 10 Nations XX Viewers

Every circuit + Sepang circuit

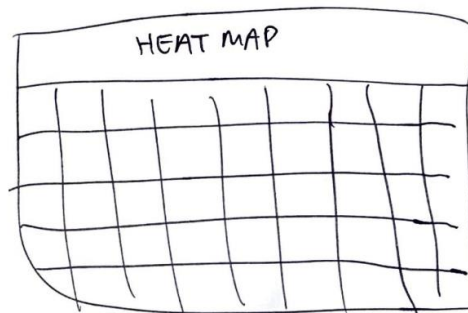
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Races by country

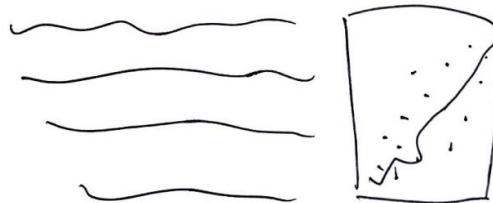
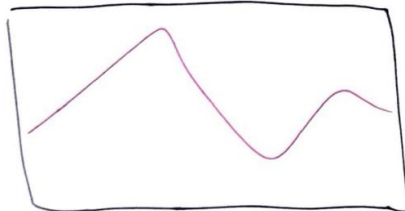


HEAT MAP

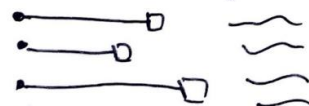


who is the best?

Speed and Strategy on Track



All races in Malaysia



Verstappen v.s. Hamilton  
Abu Dhabi 2021



Conclusion

Can Malaysia Afford to Return?

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Part D: Full-size screenshot of your DV II

Paste a full-size screenshot of your DV II assignment from the URL here



Declaration of AI Usage

I acknowledge the use of AI assistance (Claude, Anthropic) in the following specific capacities during the preparation of this assignment:

Source discovery — AI was used to suggest and locate relevant data sources, including the World Bank International Tourism Receipts dataset (ST.INT.RCPT.CD) and the DOSM Tourism Satellite Account releases. All data was independently verified, downloaded, and cited from the original sources.

Grammar and writing — AI was used to check grammar, sentence structure, and clarity of written content in the report description and visualisation annotations. All analytical arguments, design decisions, and domain knowledge are my own.

CSS styling assistance — AI was used to help write and debug custom CSS for the web page layout, including the grid system, typography hierarchy, hero section, and responsive column structure. The design decisions — colour palette, typeface selection, layout structure, and visual hierarchy — were made by me. AI assisted with translating those decisions into correct CSS syntax.

All Vega-Lite chart specifications, data transformations, idiom selections, and visualisation design choices were conceived and directed by me. AI was used as a technical assistant, not as the author of the analytical or creative work.

Sources

1. Kaggle F1 World Championship Dataset

<https://www.kaggle.com/datasets/rohanrao/formula-1-world-championship-1950-2020>

2. Formulapedia F1 Statistics <https://www.formulapedia.com/statistics/>

3. GPDestinations — Race Attendance & Ticket Prices

<https://www.gpdestinations.com/f1-ticket-prices/>

4. The World Data — F1 Revenue & TV Viewership

<https://www.theworlddata.com/formula-one/>

5. World Bank — International Tourism Receipts (ST.INT.RCPT.CD)

<https://data.worldbank.org/indicator/ST.INT.RCPT.CD?locations=MY>

6. DOSM Tourism Satellite Account 2021–2024 [https://www.dosm.gov.my/portal-](https://www.dosm.gov.my/portal-main/release-content/tourism-satellite-account-2024)

[main/release-content/tourism-satellite-account-2024](https://www.dosm.gov.my/portal-main/release-content/tourism-satellite-account-2024)

7. Wikipedia — Malaysian Grand Prix

https://en.wikipedia.org/wiki/Malaysian_Grand_Prix